GeoModRank: Self-Supervised Spatial Representation Learning for Multimodal Geospatial Data

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1 Introduction

Understanding environmental and population health outcomes requires modeling complex relationships among geography, human behavior, and environmental context [1]. Such systems are inherently spatial and multimodal, with heterogeneous coverage and cross-location dependencies. Addressing this structure requires relational inductive biases beyond independent tabular models [2, 3]. Recent geospatial foundation models, including the Population Dynamics Foundation Model (PDFM) [4], show that embeddings learned from geo-indexed signals transfer across tasks. Inspired by this paradigm, we study self-supervised spatial representation learning in a controlled synthetic setting that mirrors geospatial health data. Using masked reconstruction [5, 6], we learn graph-based embeddings [2, 3] from unlabeled multimodal features and evaluate their impact on region-level interpolation under spatial dependence, regional variation, and missing data [7, 5].

Our contributions are threefold: **(1)** a deterministic synthetic dataset with multimodal feature blocks, spatially correlated latent fields, and missingness; **(2)** GeoModRank, a modality-aware self-supervised GNN in PyTorch Geometric that incorporates spatial message passing and Laplacian regularization; and **(3)** a standardized interpolation evaluation grounded in spatial reasoning [8], comparing embedding-based predictors to coordinate-only baselines.

Although synthetic and static, this controlled environment enables systematic analysis of spatial inductive bias and modality-aware representation learning. The full pipeline is implemented in PyTorch and PyTorch Geometric [9] with fixed seeds and automated validation checks. Code and full reproducibility details are available at `https://github.com/EkleTony/healthy-climate-ssl`.

2 Synthetic Dataset Design

We construct a deterministic synthetic geospatial dataset that mimics environmental and population health data. For N regions, coordinates $s_i = (x_i, y_i)$ are sampled from a clustered Gaussian mixture to induce spatial structure:

$$s_i \sim \sum_{k=1}^{K} \pi_k \mathcal{N}(\mu_k, \Sigma_k). \tag{1}$$

A grouping variable $\texttt{state_id}_i$ introduces regional heterogeneity. Spatial structure is modeled through smooth latent fields [8, 10]. Each region includes three modalities: **(1)** climate (smooth signals), **(2)** pollution (local variability), and **(3)** socio-demographic features with group-level shifts. Structured masks simulate missing data without leakage [11]. Two regression targets depend on spatial fields, modality aggregates, and group effects, ensuring interpolation remains meaningful.

3 GeoModRank: Self-Supervised Spatial Learning

We introduce GeoModRank, a modality-aware self-supervised GNN for geospatial representation learning, illustrated in Fig. 1. The model learns region embeddings from unlabeled multimodal features by combining graph message passing [2, 3] with spatial smoothness regularization [8]. Training follows a masked reconstruction objective [5, 6], producing embeddings $z_i \in \mathbb{R}^{192}$.

Problem Setup. Let $G = (V, E)$ be a spatial k-nearest neighbor graph over $N = |V|$ regions under Euclidean distance. Each region $i \in V$ has multimodal features $x_i = \{x_i^{(c)}, x_i^{(p)}, x_i^{(s)}\}$ (climate, pollution, socio-demographic) and structured missingness represented by masks $m_i^{(m)} \in \{0, 1\}^{d_m}$. Missing entries are excluded from corruption and loss computation to prevent leakage.

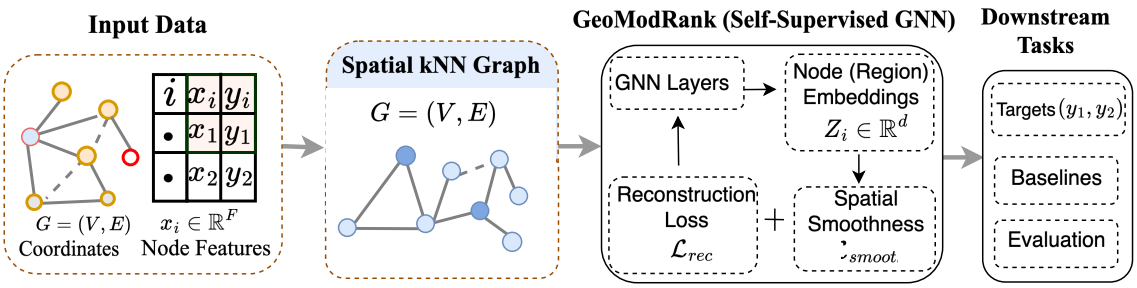

Figure 1: **GeoModRank framework.** A spatial k-nearest neighbor graph and a modality-aware self-supervised GNN produce embeddings for downstream interpolation.

Masked Self-Supervised GNN and Encoding. GeoModRank is trained via masked feature reconstruction. For observed entries, corruption masks $r_i^{(m)} \sim \text{Bernoulli}(\rho)$ are sampled with $r_{ik}^{(m)} \leq m_{ik}^{(m)}$, and corrupted inputs are $\tilde{x}_i^{(m)} = x_i^{(m)} \odot (1 - r_i^{(m)})$. Reconstruction is restricted to observed entries, consistent with masked autoencoding frameworks [5, 6]. To preserve modality structure, each feature block is encoded independently as $h_i^{(m)} = \phi_m(\tilde{x}_i^{(m)})$, where ϕ_m is a modality-specific MLP [12]. The representations are concatenated, $h_i^{(0)} = \text{concat}(h_i^{(c)}, h_i^{(p)}, h_i^{(s)})$, before mixing. Spatial inductive bias is incorporated via multi-layer GraphSAGE aggregation [3]:

$$h_i^{(\ell+1)} = \sigma\left(W^{(\ell)} \frac{1}{|\mathcal{N}(i)| + 1} \sum_{j \in \mathcal{N}(i) \cup \{i\}} h_j^{(\ell)}\right), \tag{2}$$

where $\mathcal{N}(i)$ denotes neighbors and σ is a nonlinear activation. The final embedding is $z_i = h_i^{(L)}$.

Reconstruction and Regularization. The embeddings in Eqn. 2 are partitioned by modality, $z_i = [z_i^{(c)} \| z_i^{(p)} \| z_i^{(s)}]$, and decoded $\hat{x}_i^{(m)} = \psi_m(z_i^{(m)})$. Reconstruction over masked observed entries is:

$$\mathcal{L}_{\text{recon}} = \sum_m \frac{1}{|\Omega_m|} \sum_{(i,k) \in \Omega_m} \|\hat{x}_{ik}^{(m)} - x_{ik}^{(m)}\|_2^2, \tag{3}$$

with $\Omega_m = \{(i, k) : m_{ik}^{(m)} = 1 \wedge r_{ik}^{(m)} = 1\}$. A Laplacian smoothness term $\mathcal{L}_{\text{smooth}} = \frac{1}{|E|} \sum_{(i,j) \in E} \|z_i - z_j\|_2^2$ encourages spatial coherence. The total loss is $\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}}$. Embeddings are extracted without corruption for downstream interpolation.

4 Interpolation Evaluation

We evaluate GeoModRank on a region-level interpolation task to test whether self-supervised embeddings improve prediction over coordinate-only baselines.

Data Splits and Protocol. Regions are split into train/val/test (70/10/20) with a fixed seed, stratified when applicable. Self-supervised pretraining uses all regions without labels; downstream models use training labels with validation for parameter tuning and a single test evaluation. Preprocessing is fit on training data to prevent leakage.

Coordinate Baselines. To assess how much geographic proximity explains the task, we evaluate coordinate-based predictors on standardized coordinates: **(i)** mean prediction $\hat{y}_i = \frac{1}{|\mathcal{T}|} \sum_{j \in \mathcal{T}} y_j$, **(ii)** inverse distance weighting (IDW) [13] $\hat{y}(u) = \frac{\sum_i w_i(u) y_i}{\sum_i w_i(u)}$ with $w_i(u) = \|u - u_i\|_2^{-p}$ $(p = 2)$, **(iii)** k-NN regression [14] (validation-tuned k), and **(iv)** Nadaraya–Watson kernel regression (RBF) [15].

Embedding-Based Predictors. Learned embeddings z_i are used with **(i)** Ridge regression $\hat{y}_i = w^\top z_i + b$ minimizing $\sum_{i \in \mathcal{T}} (y_i - w^\top z_i)^2 + \alpha \|w\|_2^2$, and **(ii)** a one-hidden-layer MLP $\hat{y}_i = W_2 \sigma(W_1 z_i + b_1) + b_2$ with early stopping.

Table 1: Downstream interpolation performance on the **TEST** split (seed=7). Lower RMSE is better; higher R^2 is better.

Model	y_1		y_2	
	RMSE ↓	R^2 ↑	RMSE ↓	R^2 ↑
MeanTrain	1.3739	-0.0000	0.8215	-0.0016
IDW (p=2)	0.5864	0.8178	0.5259	0.5895
kNN(coords)	0.5230	0.8551	0.5010	0.6275
NWKR (RBF coords)	0.5040	0.8654	0.4969	0.6336
GeoModRank+Ridge	**0.4915**	**0.8720**	**0.4900**	**0.6437**
GeoModRank+MLP	0.4932	0.8711	0.4924	0.6402

Metrics. Performance on held-out regions is reported using RMSE (prediction error magnitude) and R^2 (variance explained relative to the mean baseline):

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}, \quad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{4}$$

5 Results

Table 1 reports final test-set interpolation performance for y_1 and y_2. Coordinate-only baselines (IDW, kNN, NWKR) perform strongly, confirming substantial spatial smoothness in the targets. However, GeoModRank embeddings consistently improve upon the strongest baseline (NWKR).

For y_1, GeoModRank+Ridge reduces RMSE from 0.5040 to 0.4915 (**2.5**% relative reduction) and increases R^2 from 0.8654 to 0.8720 (**0.76**% relative increase). For y_2, RMSE decreases from 0.4969 to 0.4900 (**1.4**% relative reduction), with R^2 increasing from 0.6336 to 0.6437 (**1.6**% relative increase). The MLP variant performs comparably but slightly below Ridge across both targets.

Ridge achieves the best overall performance, indicating that the learned embeddings encode predictive multimodal structure in a linearly accessible form. The small margin between Ridge and MLP suggests that most useful information is already organized within the embedding space.

These results demonstrate that our proposed GeoModRank captures multimodal structure beyond geographic proximity alone, yielding consistent gains over strong spatial baselines.

6 Discussion and Extensions

The synthetic design captures key challenges of real-world environmental and population health data: spatial dependence, multimodality, heterogeneity, and incomplete coverage. However, several limitations remain. **(a)** First, the current setting is static. A natural extension would incorporate time-varying features and dynamic graphs, enabling temporal message passing or spatiotemporal transformer architectures. **(b)** Second, learned embeddings could be leveraged for causal inference tasks, for example as high-dimensional covariates for confounding adjustment, representation learning for heterogeneous treatment effects, or sensitivity analysis under spatial interference. **(c)** Finally, structured missingness mechanisms and multi-resolution hierarchies could be extended to better reflect real-world data collection processes.

Computational Resources. All experiments were conducted on macOS with a 3.2 GHz 8-core CPU and 8 GB RAM. No GPU acceleration was used. Dataset size ($N = 2000$) and embedding dimension (192) were kept modest to ensure reproducibility under limited computational resources.

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